Chapter 8. Dirichlet's Theorem

Primes in arithmetic progression via characters

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8.1 A little elementary number theory

Motivation. The arithmetic behind Dirichlet's theorem: unique factorization, and Euler's proof via $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ that $\sum 1/p$ diverges.

8.1.1 The fundamental theorem of arithmetic

$a \mid b$ means $b = ac$ for some $c \in \mathbb{Z}$. A prime is an integer $p > 1$ with only $1, p$ as positive divisors. $\gcd(a, b)$ is the largest positive common divisor; a, b are relatively prime if $\gcd(a, b) = 1$.

Theorem 1.1 (Euclid's algorithm)

For any integers a and b with $b > 0$ there exist unique integers q and r with $0 \le r < b$ such that $a = qb + r$.

Proof.

Step Existence. $\{a - qb : q \in \mathbb{Z}\} \cap \{0, 1, 2, \dots\}$ is nonempty, with least element $r = a - qb \ge 0$; $r \ge b$ would make $r - b \ge 0$ smaller. So $0 \le r < b$.

Step Uniqueness. $qb + r = q'b + r'$ gives $b \mid r - r'$ with $|r - r'| < b$, so $r = r'$, $q = q'$. ■

Theorem 1.2

If $\gcd(a, b) = d$, then there exist integers x and y such that $ax + by = d$.

Proof. Let $s = ax + by$ be the least positive integer of this form. Common divisors of a, b divide s, so $d \le s$. With $a = qs + r$, $0 \le r < s$, $r = a(1 - qx) + b(-qy)$ has the same form, so $r = 0$ and $s \mid a$; likewise $s \mid b$, so $s \le d$. ■

Corollary 1.3

Positive integers a and b are relatively prime if and only if $ax + by = 1$ for some integers x, y.

Proof. One direction is Theorem 1.2; conversely, $ax + by = 1$ means common divisors divide 1. ■

Corollary 1.4

If a and c are relatively prime and $c \mid ab$, then $c \mid b$. In particular, if a prime p does not divide a and $p \mid ab$, then $p \mid b$.

Proof. $1 = ax + cy$ gives $b = abx + cby$, divisible by c. For prime $p \nmid a$, $\gcd(a, p) = 1$. ■

Corollary 1.5

If a prime p divides $a_1 \cdots a_r$, then $p \mid a_i$ for some i.

Proof. If $p \nmid a_1$ then $p \mid a_2 \cdots a_r$ by Corollary 1.4; induct on r. ■

Theorem 1.6

Every integer greater than 1 factors uniquely into a product of primes (up to order).

Proof.

Step Existence. Otherwise take the least $n > 1$ without factorization; $n = ab$ with $1 < a, b < n$, both of which factor: a contradiction.

Step Uniqueness. If $p_1 \cdots p_r = q_1 \cdots q_s$, Corollary 1.5 gives $p_1 = q_j$ for some j; cancel and induct. ■

8.1.2 The infinitude of primes

Theorem 1.7

There are infinitely many primes.

Proof. If $p_1, \ldots, p_n$ were all the primes, a prime factor p_i of $N = p_1 \cdots p_n + 1$ would divide $N - p_1 \cdots p_n = 1$. ■

Euler's argument proves $\sum 1/p = \infty$ and is the model for Dirichlet's proof. For $s > 1$ let $\zeta(s) = \sum_{n \geq 1} n^{-s}$; comparing with $\int x^{-s} dx$,

$$\frac{1}{s-1} \leq \zeta(s) \leq 1 + \frac{1}{s-1}, \qquad \text{so } \zeta(s) \to \infty \text{ as } s \to 1^+. \tag{8.1}$$

(as $\int_n^{n+1} x^{-s} dx \leq n^{-s} \leq \int_{n-1}^{n} x^{-s} dx$). An infinite product $\prod A_n$ **converges** if $\prod_{n \leq N} A_n$ has a limit.

Lemma 1.8

The exponential and logarithm functions satisfy:
(i) $e^{\log x} = x$ for $x > 0$.
(ii) $\log(1 + x) = x + E(x)$ with $|E(x)| \leq x^2$ if $|x| < 1/2$.
(iii) If $\log(1 + x) = y$ and $|x| < 1/2$, then $|y| \leq 2|x|$.

Proof. (i) Definition of log.

(ii) For $|x| < 1$, $\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$, so $|E(x)| \leq \frac{x^2}{2(1-|x|)} \leq x^2$ for $|x| < 1/2$.

(iii) By (ii), $|y| \leq |x| + x^2 \leq |x| + \frac{1}{2}|x| \leq 2|x|$. ■

Proposition 1.9

If $A_n = 1 + a_n$ and $\sum |a_n|$ converges, then $\prod_n A_n$ converges, and the product vanishes if and only if some factor A_n vanishes. Also, if $a_n \neq 1$ for all n, then $\prod_n 1/(1 - a_n)$ converges.

Proof. For real a_n (complex: Proposition 3.2). The finitely many n with $|a_n| \geq 1/2$ give a fixed finite factor (zero iff some $A_n = 0$). For the rest, $\prod_{n \leq N} A_n = e^{B_N}$, $B_N = \sum_{n \leq N} \log(1 + a_n)$, with $|\log(1 + a_n)| \leq 2|a_n|$, so $B_N \to B$ and $\prod A_n \to e^B \neq 0$. Finally $1/(1 - a_n) = 1 + a_n/(1 - a_n)$ with $|a_n/(1 - a_n)| \leq 2|a_n|$ once $|a_n| < 1/2$. ■

Theorem 1.10

For every $s > 1$, $\zeta(s) = \displaystyle\prod_p \frac{1}{1 - p^{-s}}$, the product being over all primes.

Proof. The product converges (Proposition 1.9, $\sum_p p^{-s} < \infty$). For the primes $p_1 < \cdots < p_N$ up to M, expanding geometric series,

$$\prod_{j\le N} \frac{1}{1-p_j^{-s}} = \prod_{j\le N} \left(1 + p_j^{-s} + p_j^{-2s} + \cdots\right) = \sum_{n\in\mathcal{N}_M} n^{-s},$$

where by Theorem 1.6 $\mathcal{N}_M$ is the set of integers with all prime factors $\le M$, each once (absolutely convergent series may be multiplied and rearranged). As $\{1,\dots,M\} \subset \mathcal{N}_M$,

$$\sum_{n\le M} n^{-s} \le \prod_{p\le M} \frac{1}{1-p^{-s}} \le \zeta(s).$$

Let $M \to \infty$. ■

Proposition 1.11

The series $\sum_p 1/p$ over all primes diverges.

Proof. Take logarithms in Theorem 1.10 (the partial products tend to a positive limit); by Lemma 1.8(ii),

$$\log\zeta(s) = -\sum_p \log(1-p^{-s}) = \sum_p \left(p^{-s} + O(p^{-2s})\right) = \sum_p p^{-s} + O(1),$$

as $\sum_p p^{-2s} \le \sum n^{-2}$. As $s \to 1^+$, $\zeta(s) \to \infty$ by (8.1), so $\sum_p p^{-s} \to \infty$, and $p^{-s} \le p^{-1}$ gives $\sum_p 1/p = \infty$. ■

8.2 Dirichlet's theorem

Motivation. The indicator of the class ℓ expands in characters of $\mathbb{Z}^*(q)$, reducing the theorem to the boundedness of $\sum \chi(p)p^{-s}$ for $\chi \neq \chi_0$, i.e. to L-functions at $s = 1$.

Theorem 2.1 (Dirichlet)

If q and ℓ are relatively prime positive integers, then there are infinitely many primes of the form $\ell + kq$ with $k \in \mathbb{Z}$.

As Euler, we show $\sum_{p\equiv\ell\,(q)} p^{-s} \to \infty$ as $s \to 1^+$ (completed in Section 8.3).

8.2.1 Fourier analysis, Dirichlet characters, and reduction of the theorem

$\mathbb{Z}^*(q)$, the residues mod q prime to q, is a multiplicative group (Corollary 1.3) of order $\varphi(q)$. For the indicator δ_ℓ of ℓ, Theorem 2.7 gives

$$\delta_\ell(m) = \sum_{e\in\widehat{\mathbb{Z}^*(q)}} \hat\delta_\ell(e)e(m), \qquad \hat\delta_\ell(e) = \frac{1}{\varphi(q)} \sum_{m\in\mathbb{Z}^*(q)} \delta_\ell(m)\overline{e(m)} = \frac{1}{\varphi(q)}\overline{e(\ell)}.$$

A character e of $\mathbb{Z}^*(q)$ gives the **Dirichlet character** $\chi = \chi_e$ on $\mathbb{Z}$: $\chi(m) = e(m \bmod q)$ if $\gcd(m,q) = 1$, else 0; χ_0 comes from $e \equiv 1$. Extend δ_ℓ to $\mathbb{Z}$ by $\delta_\ell(m) = \mathbb{1}\{m \equiv \ell \pmod q\}$.

Lemma 2.2

The Dirichlet characters are multiplicative: $\chi(mn)=\chi(m)\chi(n)$ for all $m,n\in\mathbb{Z}$. Moreover

$$\delta_\ell(m)=\frac{1}{\varphi(q)}\sum_{\chi}\overline{\chi(\ell)}\chi(m),$$

summing over all Dirichlet characters mod q.

Proof. For m,n prime to q, $\chi(mn)=e(m)e(n)$; otherwise mn is not prime to q and both sides vanish. The formula holds for m prime to q by the expansion; for $\gcd(m,q)>1$ both sides are 0 ($m\not\equiv\ell$, $\chi(m)=0$). ∎

Hence, for $s>1$,

$$\sum_{p\equiv\ell\,(q)}\frac{1}{p^s}=\sum_{p}\frac{\delta_\ell(p)}{p^s}=\frac{1}{\varphi(q)}\sum_{p\nmid q}\frac{1}{p^s}+\frac{1}{\varphi(q)}\sum_{\chi\neq\chi_0}\overline{\chi(\ell)}\sum_{p}\frac{\chi(p)}{p^s}. \tag{8.2}$$

The first term tends to ∞ as $s\to1^+$ (Proposition 1.11; finitely many $p\mid q$), so Dirichlet's theorem follows from:

Theorem 2.3

If χ is a nontrivial Dirichlet character, then $\sum_p\chi(p)p^{-s}$ remains bounded as $s\to1^+$.

The proof occupies Section 8.3.

8.2.2 Dirichlet L-functions

For a Dirichlet character χ and $s>1$ define

$$L(s,\chi)=\sum_{n=1}^{\infty}\frac{\chi(n)}{n^s}.$$

Theorem 2.4 (Dirichlet product formula)

If $s>1$, then

$$\sum_{n=1}^{\infty}\frac{\chi(n)}{n^s}=\prod_{p}\frac{1}{1-\chi(p)p^{-s}}.$$

Proof. As for Theorem 1.10, with $|\chi(p)p^{-s}|\le p^{-s}$ (convergence by Proposition 3.2) and multiplicativity (Lemma 2.2): $\prod_{p\le M}(1+\chi(p)p^{-s}+\chi(p)^2p^{-2s}+\cdots)=\sum_{n\in\mathcal{N}_M}\chi(n)n^{-s}$, which differs from $\sum_n\chi(n)n^{-s}$ by at most $\sum_{n>M}n^{-s}\to0$. ∎

Formally, with $\log(1+x)=x+O(x^2)$,

$$\log L(s,\chi)=-\sum_{p}\log\left(1-\chi(p)p^{-s}\right)=\sum_{p}\frac{\chi(p)}{p^s}+O(1).$$

So if $L(1,\chi)$ is finite and nonzero, $\sum_p\chi(p)p^{-s}$ stays bounded. Rigour needs:

(a) a complex logarithm, as χ may be complex;

(b) continuity of $L(s,\chi)$ down to $s=1$; and (c) the key fact $L(1,\chi)\neq0$.

8.3 Proof of the theorem

Motivation. A power-series logarithm, continuation of $L(s,\chi)$ to $s>0$, and $L(1,\chi)\neq 0$ (positivity for complex χ, hyperbola sums for real χ) make the argument rigorous.

8.3.1 Logarithms

For $|z|<1$ define

$$\log_1\Big(\frac{1}{1-z}\Big)=\sum_{k=1}^{\infty}\frac{z^k}{k}.$$

Proposition 3.1

The function $\log_1$ satisfies:
(i) If $|z|<1$, then $e^{\log_1(\frac{1}{1-z})}=\frac{1}{1-z}$.
(ii) If $|z|<1$, then $\log_1\left(\frac{1}{1-z}\right)=z+E_1(z)$, where $|E_1(z)|\le|z|^2$ if $|z|<1/2$.
(iii) If $|z|<1/2$, then $\left|\log_1\left(\frac{1}{1-z}\right)\right|\le 2|z|$.

Proof. (i) With $z=re^{i\theta}$, show $(1-re^{i\theta})\,e^{\sum_k (re^{i\theta})^k/k}=1$. Its r-derivative is

$$\Big[-e^{i\theta}+(1-re^{i\theta})\,e^{i\theta}\sum_{k\ge 1}(re^{i\theta})^{k-1}\Big]e^{\sum_k (re^{i\theta})^k/k}=\Big[-e^{i\theta}+(1-re^{i\theta})\frac{e^{i\theta}}{1-re^{i\theta}}\Big]e^{\cdots}=0.$$

so it is constant in r, equal to 1 at $r=0$.

(ii) $|E_1(z)|=\left|\sum_{k\ge 2}z^k/k\right|\le\frac{|z|^2}{2}\sum_{j\ge 0}|z|^j=\frac{|z|^2}{2(1-|z|)}\le|z|^2$ for $|z|<1/2$.

(iii) By (ii), $|\log_1(\frac{1}{1-z})|\le|z|+|z|^2\le 2|z|$. ∎

Proposition 3.2

If $\sum|a_n|$ converges and $a_n\neq 1$ for all n, then $\prod_{n=1}^{\infty}\frac{1}{1-a_n}$ converges and is nonzero.

Proof. As in Proposition 1.9: $|a_n|<1/2$ for $n\ge n_0$, and by Proposition 3.1(i),

$$\prod_{n_0\le n\le N}\frac{1}{1-a_n}=\exp\Big(\sum_{n_0\le n\le N}\log_1\Big(\frac{1}{1-a_n}\Big)\Big).$$

The exponent converges absolutely by (iii), so the products tend to $e^B\neq 0$; the finitely many factors $n<n_0$ are finite and nonzero. ∎

8.3.2 L-functions

Proposition 3.3

Let χ_0 be the trivial Dirichlet character ($\chi_0(n)=1$ if $\gcd(n,q)=1$, 0 otherwise) and $q=p_1^{a_1}\cdots p_N^{a_N}$. Then

$$L(s,\chi_0)=(1-p_1^{-s})(1-p_2^{-s})\cdots(1-p_N^{-s})\,\zeta(s).$$

Therefore $L(s,\chi_0)\to\infty$ as $s\to 1^+$.

Proof. By Theorems 2.4 and 1.10, $L(s,\chi_0)=\prod_{p\nmid q}(1-p^{-s})^{-1}$ is the Euler product of ζ without the factors for $p_1,\dots,p_N$; then use (8.1). ∎

Proposition 3.4

If χ is a nontrivial Dirichlet character, then $\sum_{n\ge1}\chi(n)/n^s$ converges for $s>0$; we denote its sum by $L(s,\chi)$. Moreover:
(i) $L(s,\chi)$ is continuously differentiable for $0<s<\infty$.
(ii) There are constants $c,c'>0$ with $L(s,\chi)=1+O(e^{-cs})$ and $L'(s,\chi)=O(e^{-c's})$ as $s\to\infty$.

Lemma 3.5

If χ is a nontrivial Dirichlet character, then $\left|\sum_{n=1}^{k}\chi(n)\right|\le q$ for all k.

Proof. $S=\sum_{n=1}^{q}\chi(n)=0$: for $a\in\mathbb{Z}^*(q)$, $\chi(a)S=\sum_n\chi(an)=S$ ($n\mapsto an$ permutes residues), and $\chi(a)\neq1$ for some a. By q-periodicity, with $k=aq+b$, $0\le b<q$, $\sum_{n=1}^{k}\chi(n)$ has at most q remaining terms of modulus ≤1. ∎

Proof of Proposition 3.4. Let $s_k=\sum_{n=1}^{k}\chi(n)$, $s_0=0$, $|s_k|\le q$. Summation by parts:

$$\sum_{n=1}^{N}\frac{\chi(n)}{n^s}=\sum_{n=1}^{N}\frac{s_n-s_{n-1}}{n^s}=\sum_{n=1}^{N-1}s_n\Big[\frac{1}{n^s}-\frac{1}{(n+1)^s}\Big]+\frac{s_N}{N^s}.$$

By the mean value theorem $|n^{-s}-(n+1)^{-s}|\le s\,n^{-s-1}$. For $s\ge\delta>0$ the boundary term tends to 0 and the sum converges absolutely and uniformly on compacts of $[\delta,\infty)$, as $|s_n|\,s\,n^{-s-1}\le qs\,n^{-\delta-1}$. So $L(s,\chi)=\sum_n s_n[n^{-s}-(n+1)^{-s}]$ is continuous for $s>0$.

(i) The termwise derivative $\sum_n s_n[-\log n\,n^{-s}+\log(n+1)(n+1)^{-s}]$ converges uniformly on compacts too, since $g(x)=x^{-s}\log x$ has $|g'(x)|\le C_\delta x^{-1-\delta/2}$ for $s\ge\delta$. So $L(\cdot,\chi)\in C^1(0,\infty)$.

(ii) $\chi(1)=1$, so $|L(s,\chi)-1|\le\sum_{n\ge2}n^{-s}\le 2^{-s}(1+\frac{2}{s-1})=O(e^{-cs})$ for $s\ge2$, any $c<\log2$; likewise $|L'(s,\chi)|\le\sum_{n\ge2}(\log n)n^{-s}=O(e^{-c's})$. ∎

For nontrivial χ and $s>1$ ($L(s,\chi)\neq0$ by Theorem 2.4 and Proposition 3.2), as $L(s,\chi)\to1$ for $s\to\infty$, define

$$\log_2 L(s,\chi)=-\int_s^\infty\frac{L'(t,\chi)}{L(t,\chi)}\,dt,$$

convergent as $L'/L=O(e^{-c't})$ ($\log_2$ is a name, not base 2). If $L(1,\chi)\neq0$ the integral extends continuously to $s=1$, so $\log_2 L(s,\chi)$ stays bounded as $s\to1^+$.

Proposition 3.6

If $s>1$, then $e^{\log_2 L(s,\chi)}=L(s,\chi)$. Moreover

$$\log_2 L(s,\chi)=\sum_p\log_1\Big(\frac{1}{1-\chi(p)/p^s}\Big).$$

Proof.

Step First identity. $\frac{d}{ds}(e^{-\log_2 L}L)=e^{-\log_2 L}(-\frac{L'}{L}L+L')=0$, and as $s\to\infty$, $\log_2 L\to0$, $L\to1$; so the constant is 1.

Step Second identity. By Proposition 3.1(i) and Theorem 2.4,

$$e^{\sum_p\log_1(\frac{1}{1-\chi(p)p^{-s}})}=\prod_p\frac{1}{1-\chi(p)p^{-s}}=L(s,\chi).$$

So the two sides differ by $2\pi i M(s)$, $M(s) \in \mathbb{Z}$. Both are continuous for $s > 1$ (uniform convergence on $[\delta, \infty)$, $\delta > 1$, by Proposition 3.1(iii)), so M is constant, and both tend to 0 as $s \to \infty$: $M \equiv 0$. ■

By Proposition 3.1(ii),

$$\log_2 L(s,\chi) = \sum_p \log_1 \Big(\frac{1}{1-\chi(p)/p^s}\Big) = \sum_p \frac{\chi(p)}{p^s} + O\Big(\sum_p \frac{1}{p^{2s}}\Big) = \sum_p \frac{\chi(p)}{p^s} + O(1).$$

So $L(1,\chi) \neq 0$ makes $\sum_p \chi(p)p^{-s}$ bounded as $s \to 1^+$, and Theorem 2.3 (hence Dirichlet's theorem) reduces to:

8.3.3 Non-vanishing of the L-function

Theorem 3.7

If $\chi \neq \chi_0$, then $L(1,\chi) \neq 0$.

A Dirichlet character is **real** if it takes only real values $(0, \pm 1)$, i.e. $\chi = \bar{\chi}$, and **complex** otherwise.

Case I: complex Dirichlet characters

Lemma 3.8

If $s > 1$, then $\prod_\chi L(s,\chi) \geq 1$, the product being over all Dirichlet characters mod q. In particular the product is real-valued.

Proof. By Proposition 3.6 (and its analogue for χ_0 from the Euler product), $L(s,\chi) = \exp\big(\sum_p \log_1 \frac{1}{1-\chi(p)p^{-s}}\big)$, so

$$\prod_\chi L(s,\chi) = \exp\Big(\sum_\chi \sum_p \sum_{k\geq 1} \frac{1}{k}\frac{\chi(p^k)}{p^{ks}}\Big) = \exp\Big(\sum_p \sum_{k\geq 1} \frac{1}{kp^{ks}} \sum_\chi \chi(p^k)\Big),$$

using $\chi(p)^k = \chi(p^k)$ and absolute convergence. By Lemma 2.2 with $\ell = 1$, $\sum_\chi \chi(p^k) = \varphi(q)\delta_1(p^k) \geq 0$, so the exponent is ≥ 0. ■

Lemma 3.9

The following hold:
(i) If $L(1,\chi) = 0$, then $L(1,\bar{\chi}) = 0$.
(ii) If χ is nontrivial and $L(1,\chi) = 0$, then $|L(s,\chi)| \leq C|s-1|$ for $1 \leq s \leq 2$.
(iii) For the trivial character, $|L(s,\chi_0)| \leq \frac{C}{|s-1|}$ for $1 < s \leq 2$.

Proof. (i) $L(1,\bar{\chi}) = \overline{L(1,\chi)}$ (real n^{-s}).

(ii) Mean value theorem, $L(\cdot,\chi) \in C^1(0,\infty)$ (Proposition 3.4): $|L(s,\chi) - L(1,\chi)| \leq \sup_{[1,2]} |L'| \cdot |s-1|$.

(iii) By Proposition 3.3 and (8.1), $|L(s,\chi_0)| \leq \zeta(s) \leq \frac{2}{s-1}$. ■

Proof of Theorem 3.7 for complex χ. If $L(1,\chi) = 0$, then $L(1,\bar{\chi}) = 0$ with $\bar{\chi} \neq \chi$, so in $\prod_{\chi'} L(s,\chi')$ two factors are $O(|s-1|)$, the χ_0 factor is $O(1/|s-1|)$, and the rest are bounded (continuous on $[1,2]$) (Lemma 3.9). So the product tends to 0, contradicting Lemma 3.8. ■

Case II: real Dirichlet characters

For real χ only one factor could vanish, and the pole of $L(s,\chi_0)$ can cancel it; instead we use Dirichlet's *summation along hyperbolas*, first on the divisor problem.

Sums vs. integrals.

Proposition 3.10

If N is a positive integer, then:

(i) $\displaystyle\sum_{1\le n\le N}\frac{1}{n}=\int_1^N\frac{dx}{x}+O(1)=\log N+O(1).$

(ii) More precisely, there is a real number γ, **Euler's constant**, with $\displaystyle\sum_{1\le n\le N}\frac{1}{n}=\log N+\gamma+O(1/N)$.

Proof. Prove (ii). Let $\gamma_n=\frac{1}{n}-\int_n^{n+1}\frac{dx}{x}$; as $1/x$ decreases, $0\le\gamma_n\le\frac{1}{n}-\frac{1}{n+1}\le\frac{1}{n^2}$, so $\sum\gamma_n=\gamma$ converges with $\sum_{n>N}\gamma_n\le\frac{1}{N}$. Then

$$\sum_{n=1}^{N}\frac{1}{n}-\int_1^N\frac{dx}{x}=\sum_{n=1}^{N}\gamma_n+\int_N^{N+1}\frac{dx}{x}=\gamma-\sum_{n>N}\gamma_n+\int_N^{N+1}\frac{dx}{x},$$

with the last two terms $O(1/N)$. ∎

Proposition 3.11

If N is a positive integer, then

$$\sum_{1\le n\le N}\frac{1}{n^{1/2}}=\int_1^N\frac{dx}{x^{1/2}}+c'+O(N^{-1/2})=2N^{1/2}+c+O(N^{-1/2}).$$

Proof. [Supplement] Let $\gamma_n=n^{-1/2}-\int_n^{n+1}x^{-1/2}dx$; by the mean value theorem $0\le\gamma_n\le\frac{1}{2}n^{-3/2}$, so $\sum\gamma_n=c'$ with $\sum_{n>N}\gamma_n=O(N^{-1/2})$. As before,

$$\sum_{n\le N}n^{-1/2}-\int_1^N x^{-1/2}dx=c'-\sum_{n>N}\gamma_n+\int_N^{N+1}x^{-1/2}dx=c'+O(N^{-1/2}),$$

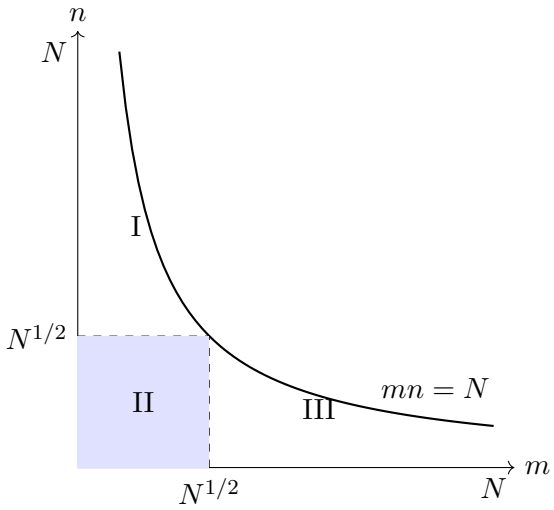
and $\int_1^N x^{-1/2}dx=2N^{1/2}-2$; take $c=c'-2$. ∎

Hyperbolic sums. For F on pairs of positive integers, $S_N=\sum\sum_{mn\le N}F(m,n)$ can be computed

(a) along hyperbolas: $S_N=\sum_{1\le k\le N}\big(\sum_{nm=k}F(m,n)\big)$;

(b) vertically: $S_N=\sum_{1\le m\le N}\big(\sum_{1\le n\le N/m}F(m,n)\big)$;

(c) horizontally: $S_N=\sum_{1\le n\le N}\big(\sum_{1\le m\le N/n}F(m,n)\big)$.



Split $\{mn \le N\}$ into $\mathrm{I} = \{1 \le m < N^{1/2},\ N^{1/2} < n \le N/m\}$, $\mathrm{II} = \{1 \le m \le N^{1/2},\ 1 \le n \le N^{1/2}\}$, and $\mathrm{III} = \{N^{1/2} < m \le N/n,\ 1 \le n < N^{1/2}\}$, with corresponding sums $S_{\mathrm{I}}, S_{\mathrm{II}}, S_{\mathrm{III}}$.

Let $d(k) = \sum_{nm=k} 1$, the number of divisors of k.

Theorem 3.12 (Dirichlet's divisor theorem)

For positive integers N,

$$\frac{1}{N}\sum_{k=1}^{N} d(k) = \log N + O(1).$$

More precisely, $\frac{1}{N}\sum_{k=1}^{N} d(k) = \log N + (2\gamma - 1) + O(N^{-1/2})$.

Proof. Let $S_N = \sum_{k \le N} d(k)$, the hyperbolic sum of $F \equiv 1$.

Step First estimate. Vertically, $S_N = \sum_{m \le N}[N/m] = N\sum_{m \le N}\frac{1}{m} + O(N)$, so $S_N/N = \log N + O(1)$ by Proposition 3.10(i).

Step Refined estimate. By symmetry $S_{\mathrm{I}} = S_{\mathrm{III}}$, so $S_N = 2(S_{\mathrm{I}} + S_{\mathrm{II}}) - S_{\mathrm{II}}$, and by Proposition 3.10(ii)

$$S_{\mathrm{I}} + S_{\mathrm{II}} = \sum_{m \le N^{1/2}} [N/m] = N \sum_{m \le N^{1/2}} \frac{1}{m} + O(N^{1/2}) = N \log N^{1/2} + N\gamma + O(N^{1/2}).$$

With $S_{\mathrm{II}} = [N^{1/2}]^2 = N + O(N^{1/2})$, $S_N = N \log N + (2\gamma - 1)N + O(N^{1/2})$. ∎

Non-vanishing for real characters. For a nontrivial real χ let $F(m,n) = \dfrac{\chi(n)}{(nm)^{1/2}}$ and $S_N = \sum\sum_{mn \le N} F(m,n)$.

Proposition 3.13

The following hold:
(i) $S_N \ge c \log N$ for some constant $c > 0$.
(ii) $S_N = 2N^{1/2}L(1,\chi) + O(1)$.

This gives Theorem 3.7 for real χ: $L(1,\chi) = 0$ would make $S_N = O(1)$, contradicting (i).

Along hyperbolas, $\sum_{nm=k} \frac{\chi(n)}{(nm)^{1/2}} = \frac{1}{k^{1/2}} \sum_{n|k} \chi(n)$, so (i) follows from:

Lemma 3.14

For every $k \geq 1$, $\sum_{n|k} \chi(n) \geq 0$, and $\sum_{n|k} \chi(n) \geq 1$ if $k = \ell^2$ for some $\ell \in \mathbb{Z}$.

Proof.

Step Prime powers. For $k = p^a$, $\sum_{n|k} \chi(n) = 1 + \chi(p) + \cdots + \chi(p)^a$ with $\chi(p) \in \{0, \pm 1\}$: $a+1$ if $\chi(p) = 1$; 1 or 0 if $\chi(p) = -1$ and a is even or odd; 1 if $\chi(p) = 0$. Always ≥ 0, and ≥ 1 for even a.

Step General k. For $k = p_1^{a_1} \cdots p_N^{a_N}$, by multiplicativity

$$\sum_{n|k} \chi(n) = \prod_{j=1}^{N} (1 + \chi(p_j) + \cdots + \chi(p_j^{a_j})).$$

Each factor is ≥ 0, and ≥ 1 for squares (all a_j even). ■

Proof of Proposition 3.13(i). By the lemma, $S_N \geq \sum_{\ell \leq N^{1/2}} \frac{1}{(\ell^2)^{1/2}} = \sum_{\ell \leq N^{1/2}} \frac{1}{\ell} \geq c \log N$ (Proposition 3.10(i)). ■

Lemma 3.15

For all integers $0 < a < b$:

(i) $\displaystyle\sum_{n=a}^{b} \frac{\chi(n)}{n^{1/2}} = O(a^{-1/2})$;

(ii) $\displaystyle\sum_{n=a}^{b} \frac{\chi(n)}{n} = O(a^{-1})$.

Proof. Summation by parts as in Proposition 3.4, $|s_n| \leq q$ (Lemma 3.5):

$$\sum_{n=a}^{b} \frac{\chi(n)}{n^{1/2}} = \sum_{n=a}^{b-1} s_n [n^{-1/2} - (n+1)^{-1/2}] + \frac{s_b}{b^{1/2}} - \frac{s_{a-1}}{a^{1/2}},$$

$$\Big|\sum_{n=a}^{b} \frac{\chi(n)}{n^{1/2}}\Big| \leq q \sum_{n=a}^{b-1} [n^{-1/2} - (n+1)^{-1/2}] + \frac{2q}{a^{1/2}} \leq \frac{3q}{a^{1/2}},$$

by telescoping; (ii) likewise with n^{-1}, bound $3q/a$. ■

Proof of Proposition 3.13(ii). Write $S_N = S_{\mathrm{I}} + (S_{\mathrm{II}} + S_{\mathrm{III}})$.

Step Region I, summed vertically.

$$S_{\mathrm{I}} = \sum_{m < N^{1/2}} \frac{1}{m^{1/2}} \Big(\sum_{N^{1/2} < n \leq N/m} \frac{\chi(n)}{n^{1/2}} \Big).$$

The inner sum is $O(N^{-1/4})$ (Lemma 3.15(i)) and $\sum_{m<N^{1/2}} m^{-1/2} = O(N^{1/4})$ (Proposition 3.11), so $S_{\mathrm{I}} = O(1)$.

Step Regions II and III, summed horizontally.

$$S_{\mathrm{II}} + S_{\mathrm{III}} = \sum_{n \leq N^{1/2}} \frac{\chi(n)}{n^{1/2}} \Big(\sum_{m \leq N/n} \frac{1}{m^{1/2}} \Big) = \sum_{n \leq N^{1/2}} \frac{\chi(n)}{n^{1/2}} \Big(2\frac{N^{1/2}}{n^{1/2}} + c + O\Big(\frac{n^{1/2}}{N^{1/2}}\Big) \Big)$$

by Proposition 3.11, i.e.

$$2N^{1/2} \sum_{n \leq N^{1/2}} \frac{\chi(n)}{n} + c \sum_{n \leq N^{1/2}} \frac{\chi(n)}{n^{1/2}} + O\Big(\frac{1}{N^{1/2}} \sum_{n \leq N^{1/2}} 1 \Big).$$

The third is $O(1)$, the second $O(1)$ by Lemma 3.15(i), and Lemma 3.15(ii) gives $\sum_{n\le N^{1/2}} \frac{\chi(n)}{n} = L(1,\chi) + O(N^{-1/2})$. So $S_N = 2N^{1/2}L(1,\chi) + O(1)$. ■

This proves Theorem 3.7, hence Theorem 2.3, and by (8.2) $\sum_{p\equiv \ell\,(q)} p^{-s} \to \infty$ as $s \to 1^+$: infinitely many primes $p \equiv \ell \pmod q$ (Theorem 2.1).